

XRD Compound Identification Report

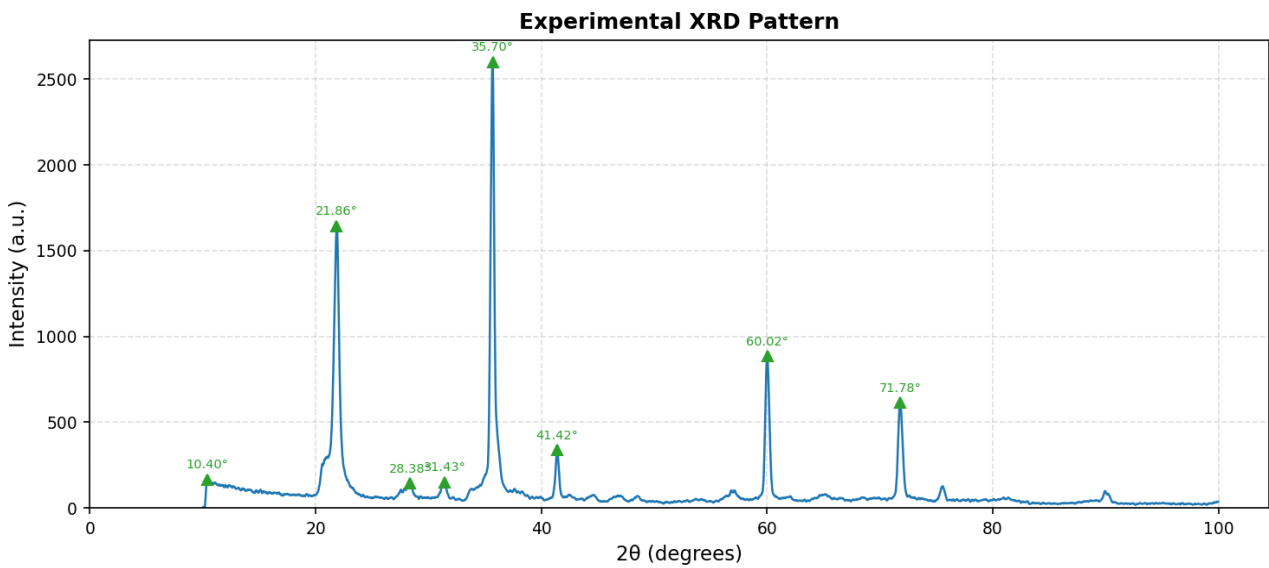
File ID: 9428008f-372b-448b-b459-c5196eeca13

Analysis Summary

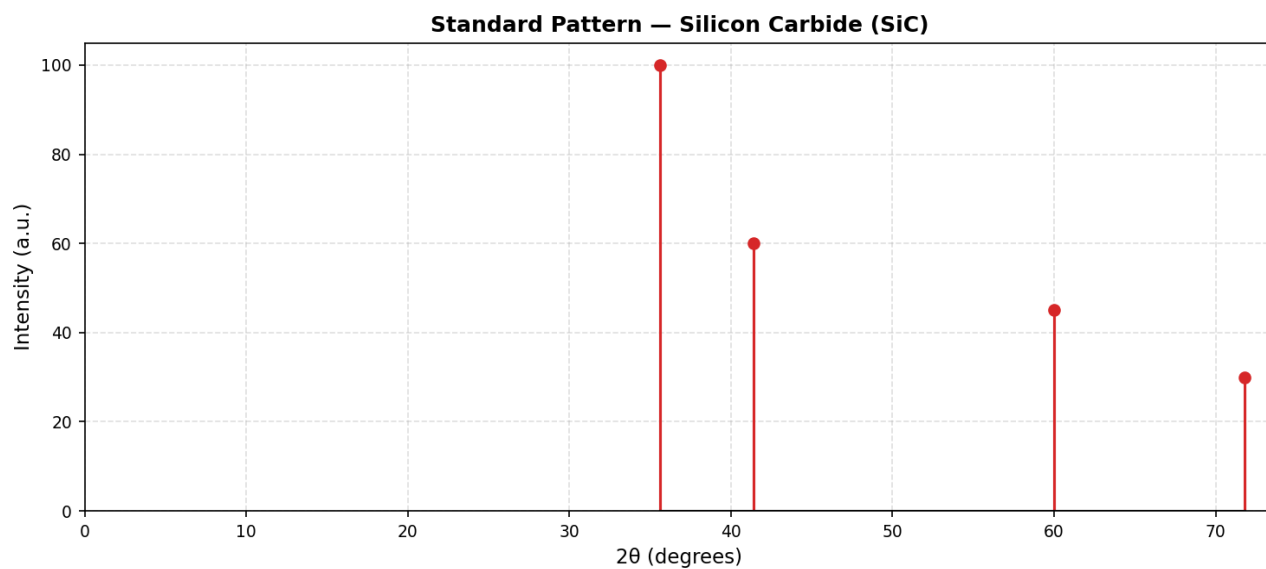
Compound Identified	Silicon Carbide
Chemical Formula	SiC
Crystal System	Hexagonal
Space Group	P63mc
Confidence Score	57.1 %
Crystallite Size	19.2 nm (192.5 Å)
Mean Peak Shift	+0.0325°
Strain Indicator	None
Detected Phases	SiC

XRD Patterns

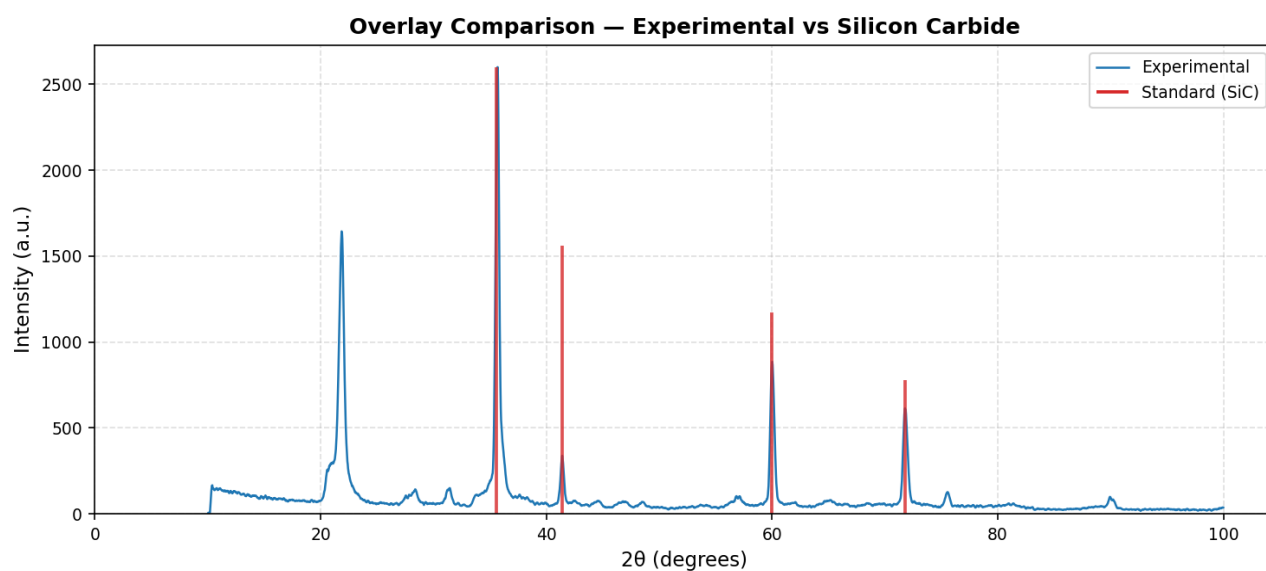
Experimental XRD Pattern



Standard Reference Pattern



Overlay Comparison



Peak Match Table

2θ exp (°)	2θ std (°)	Δ2θ (°)	d (Å)	I(rel.)	h	k	l	Match
35.695	35.600	+0.0950	2.5200	100	1	0	1	✓
41.425	41.400	+0.0250	2.1700	60	1	0	2	✓
60.025	60.000	+0.0250	1.5400	45	1	1	1	✓
71.785	71.800	+0.0150	1.3200	30	2	0	0	✓

Analysis Results

Parameter	Value
Crystallite Size (Scherrer)	19.25 nm
Crystallite Size (Ångströms)	192.52 Å
Mean Peak Shift (Δ2θ)	+0.0325°
Strain Indicator	None
Confidence Score	57.1 %
Detected Phases	SiC

Crystal Information

Property	Value
Compound	Silicon Carbide
Formula	SiC
Crystal System	Hexagonal
Space Group	P63mc